Supplementary Document for the Paper: Negative Learning Ant Colony Optimization for MaxSAT

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Table 1: Average results of all algorithms tested on Villagra and Baran's instances

	Instance	n_l	n_x	n_c	ACO$_{\text{SAW}}$	ACO$_{\text{RF}}$	ACO$_{\text{RFSAW}}$	*Walk*SAT	CPLEX	ACO$^{+}_{\text{neg}}$	ACO$_{\text{neg}}$	ACO
1	VB-3SAT-V50-C215-1	3	50	215	211.00	212.40	210.80	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
2	VB-3SAT-V50-C215-2	3	50	215	212.40	212.40	212.10	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
3	VB-3SAT-V50-C215-3	3	50	215	212.50	213.20	214.00	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
4	VB-3SAT-V50-C215-4	3	50	215	211.80	212.60	212.50	**215.00**	**215.00**	**215.00**	**215.00**	214.90
5	VB-3SAT-V50-C215-5	3	50	215	209.60	211.90	208.60	**214.00**	**214.00**	**214.00**	**214.00**	**214.00**
6	VB-3SAT-V50-C215-6	3	50	215	207.20	210.10	208.90	**214.00**	**214.00**	**214.00**	**214.00**	**214.00**
7	VB-3SAT-V50-C215-7	3	50	215	209.80	211.20	210.30	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
8	VB-3SAT-V50-C215-8	3	50	215	209.20	212.60	211.00	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
9	VB-3SAT-V50-C215-9	3	50	215	207.20	211.70	210.60	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
10	VB-3SAT-V50-C215-10	3	50	215	210.80	211.70	212.10	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
11	VB-3SAT-V50-C215-11	3	50	215	212.20	213.50	212.10	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
12	VB-3SAT-V50-C215-12	3	50	215	209.20	211.30	208.90	**214.00**	**214.00**	**214.00**	**214.00**	**214.00**
13	VB-3SAT-V50-C215-13	3	50	215	211.00	211.80	211.10	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
14	VB-3SAT-V50-C215-14	3	50	215	211.30	212.10	211.00	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
15	VB-3SAT-V50-C215-15	3	50	215	212.50	213.20	212.90	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
16	VB-3SAT-V50-C215-16	3	50	215	210.30	212.10	209.80	**214.00**	**214.00**	**214.00**	**214.00**	**214.00**
17	VB-3SAT-V50-C215-17	3	50	215	211.00	211.50	211.30	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
18	VB-3SAT-V50-C215-18	3	50	215	210.60	212.70	211.30	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
19	VB-3SAT-V50-C215-19	3	50	215	213.20	213.40	212.30	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
20	VB-3SAT-V50-C215-20	3	50	215	210.70	212.70	211.90	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
21	VB-3SAT-V50-C215-21	3	50	215	208.00	213.00	211.60	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
22	VB-3SAT-V50-C215-22	3	50	215	210.40	211.00	208.90	**215.00**	**215.00**	**215.00**	**215.00**	**215.00**
23	VB-3SAT-V50-C215-23	3	50	215	212.30	213.20	212.40	**214.00**	**214.00**	**214.00**	**214.00**	**214.00**
24	VB-3SAT-V50-C215-24	3	50	215	211.30	212.90	212.10	**214.00**	**214.00**	**214.00**	**214.00**	**214.00**
25	VB-3SAT-V50-C215-25	3	50	215	210.90	211.90	211.10	**214.00**	**214.00**	**214.00**	**214.00**	**214.00**

Continuation of Table 1 Average results of all algorithms tested on Villagra and Baran's instances

	Instance	n_l	n_x	n_c	ACO$_{\text{SAW}}$	ACO$_{\text{RF}}$	ACO$_{\text{RFSAW}}$	*Walk*SAT	CPLEX	ACO$^{+}_{\text{neg}}$	ACO$_{\text{neg}}$	ACO
26	VB-3SAT-V50-C323-1	3	50	323	304.40	313.80	304.20	**317.00**	**317.00**	**317.00**	**317.00**	**317.00**
27	VB-3SAT-V50-C323-2	3	50	323	307.10	314.30	309.90	**318.00**	**318.00**	**318.00**	317.90	317.30
28	VB-3SAT-V50-C323-3	3	50	323	308.40	314.60	305.60	**317.00**	**317.00**	**317.00**	**317.00**	**317.00**
29	VB-3SAT-V50-C323-4	3	50	323	312.00	315.70	313.20	**318.00**	**318.00**	**318.00**	**318.00**	317.90
30	VB-3SAT-V50-C323-5	3	50	323	303.00	312.80	301.10	**316.00**	**316.00**	**316.00**	**316.00**	**316.00**
31	VB-3SAT-V50-C323-6	3	50	323	307.40	315.20	308.00	**318.00**	**318.00**	**318.00**	**318.00**	317.30
32	VB-3SAT-V50-C323-7	3	50	323	305.20	314.60	306.00	**318.00**	**318.00**	**318.00**	**318.00**	**318.00**
33	VB-3SAT-V50-C323-8	3	50	323	305.30	314.50	305.50	**317.00**	**317.00**	**317.00**	**317.00**	**317.00**
34	VB-3SAT-V50-C323-9	3	50	323	304.20	312.40	304.80	**316.00**	**316.00**	**316.00**	**316.00**	**316.00**
35	VB-3SAT-V50-C323-10	3	50	323	307.00	313.80	307.80	**318.00**	**318.00**	**318.00**	**318.00**	317.10
36	VB-3SAT-V50-C323-11	3	50	323	308.00	314.90	305.40	**318.00**	**318.00**	**318.00**	**318.00**	**318.00**
37	VB-3SAT-V50-C323-12	3	50	323	304.50	313.80	305.60	**317.00**	**317.00**	**317.00**	**317.00**	**317.00**
38	VB-3SAT-V50-C323-13	3	50	323	305.30	315.60	310.70	**318.00**	**318.00**	**318.00**	**318.00**	317.80
39	VB-3SAT-V50-C323-14	3	50	323	312.60	316.00	313.30	**320.00**	**320.00**	**320.00**	**320.00**	**320.00**
40	VB-3SAT-V50-C323-15	3	50	323	307.90	313.90	307.30	**317.00**	**317.00**	**317.00**	**317.00**	**317.00**
41	VB-3SAT-V50-C323-16	3	50	323	311.30	314.80	311.10	**319.00**	**319.00**	**319.00**	**319.00**	**319.00**
42	VB-3SAT-V50-C323-17	3	50	323	311.40	318.20	313.40	**320.00**	**320.00**	**320.00**	**320.00**	**320.00**
43	VB-3SAT-V50-C323-18	3	50	323	301.10	311.90	300.80	**316.00**	**316.00**	**316.00**	**316.00**	315.90
44	VB-3SAT-V50-C323-19	3	50	323	303.50	312.20	305.90	**316.00**	**316.00**	**316.00**	**316.00**	**316.00**
45	VB-3SAT-V50-C323-20	3	50	323	305.60	313.70	305.30	**317.00**	**317.00**	**317.00**	**317.00**	**317.00**
46	VB-3SAT-V50-C323-21	3	50	323	303.20	312.40	303.70	**317.00**	**317.00**	**317.00**	316.80	316.50
47	VB-3SAT-V50-C323-22	3	50	323	304.60	313.10	304.60	**316.00**	**316.00**	**316.00**	**316.00**	**316.00**
48	VB-3SAT-V50-C323-23	3	50	323	302.60	311.50	303.60	**316.00**	**316.00**	**316.00**	314.80	314.60
49	VB-3SAT-V50-C323-24	3	50	323	305.50	313.90	307.20	**318.00**	**318.00**	**318.00**	**318.00**	317.20
50	VB-3SAT-V50-C323-25	3	50	323	309.80	314.80	310.10	**318.00**	**318.00**	**318.00**	**318.00**	**318.00**

Table 2: Average results of all algorithms tested on MSE 2020 and 2016 instances

	Instance	Group	n_l	n_x	n_c	SAT*Like*	SlsMcs	Cplex	$\text{AcoSat}^{+}_{\text{neg}}$	$\text{AcoSls}^{+}_{\text{neg}}$	$\text{Aco}^{+}_{\text{neg}}$	Aco_{neg}	Aco
1	san400_0.7_3.clq	$maxcut_1$	2	40	1094	**230.0**	**230.0**	232.0	**230.0**	**230.0**	230.3	247.7	240.7
2	san200_0.9_1.clq	$maxcut_1$	2	40	1392	**313.0**	**313.0**	317.0	**313.0**	**313.0**	315.6	328.5	321.3
3	p_hat500-1.clq	$maxcut_1$	2	42	474	**75.0**	**75.0**	**75.0**	**75.0**	**75.0**	**75.0**	86.4	82.3
4	p_hat500-3.clq	$maxcut_1$	2	42	1310	**284.0**	**284.0**	289.0	**284.0**	**284.0**	285.4	301.5	295.2
5	maxcut-140-630-0.8-3	$maxcut_2$	2	140	1258	**165.0**	**165.0**	168.0	**165.0**	**165.0**	165.6	240.4	240.1
6	maxcut-140-630-0.8-4	$maxcut_2$	2	140	1258	**165.0**	**165.0**	170.0	**165.0**	**165.0**	165.6	238.7	238.5
7	maxcut-140-630-0.8-20	$maxcut_2$	2	140	1258	**165.0**	**165.0**	**165.0**	**165.0**	**165.0**	166.3	240.1	236.4
8	maxcut-140-630-0.8-44	$maxcut_2$	2	140	1258	**160.0**	**160.0**	162.0	**160.0**	**160.0**	**160.0**	238.8	234.7
9	maxcut-140-630-0.7-3	$maxcut_2$	2	140	1260	**168.0**	169.0	173.0	**168.0**	**168.0**	170.5	240.9	236.9
10	maxcut-140-630-0.7-33	$maxcut_2$	2	140	1260	**165.0**	**165.0**	169.0	**165.0**	**165.0**	166.6	237.7	236.5
11	maxcut-140-630-0.7-49	$maxcut_2$	2	140	1260	**164.0**	169.0	170.0	**164.0**	**164.0**	165.7	238.8	237.1
12	HG-3SAT-V250-C1000-1	*highgirth*	3	250	1000	**5.0**	13.0	12.0	**5.0**	8.0	6.7	7.9	16.1
13	HG-3SAT-V250-C1000-2	*highgirth*	3	250	1000	**5.0**	11.0	13.0	**5.0**	8.6	6.7	8.7	19.4
14	HG-3SAT-V250-C1000-3	*highgirth*	3	250	1000	**5.0**	12.0	13.0	**5.0**	8.6	7.8	10.0	18.9
15	HG-3SAT-V250-C1000-4	*highgirth*	3	250	1000	**6.0**	11.0	13.0	**6.0**	9.8	8.5	10.4	17.9
16	HG-3SAT-V250-C1000-5	*highgirth*	3	250	1000	**6.0**	13.0	13.0	**6.0**	9.4	7.7	9.0	20.1
17	HG-3SAT-V250-C1000-6	*highgirth*	3	250	1000	**6.0**	14.0	10.0	**6.0**	8.8	7.4	9.0	16.9
18	HG-3SAT-V250-C1000-7	*highgirth*	3	250	1000	**6.0**	8.0	11.0	**6.0**	9.3	7.3	8.9	18.1
19	HG-3SAT-V250-C1000-8	*highgirth*	3	250	1000	**5.0**	11.0	14.0	5.9	8.7	7.5	9.0	18.7
20	HG-3SAT-V250-C1000-9	*highgirth*	3	250	1000	**6.0**	13.0	14.0	**6.0**	8.9	8.0	9.5	16.7
21	HG-3SAT-V250-C1000-10	*highgirth*	3	250	1000	**6.0**	15.0	10.0	**6.0**	8.1	7.3	8.8	17.5
22	HG-3SAT-V250-C1000-11	*highgirth*	3	250	1000	**6.0**	9.0	13.0	**6.0**	9.4	7.7	9.1	18.6
23	HG-3SAT-V250-C1000-12	*highgirth*	3	250	1000	**6.0**	10.0	16.0	**6.0**	9.3	7.1	9.6	17.6
24	HG-3SAT-V250-C1000-13	*highgirth*	3	250	1000	**5.0**	9.0	12.0	6.0	9.4	6.5	8.9	18.4
25	HG-3SAT-V250-C1000-14	*highgirth*	3	250	1000	6.0	10.0	14.0	**5.9**	7.5	6.8	7.6	18.2
26	HG-3SAT-V250-C1000-15	*highgirth*	3	250	1000	**5.0**	10.0	12.0	**5.0**	7.9	6.6	8.7	19.7
27	HG-3SAT-V250-C1000-16	*highgirth*	3	250	1000	**5.0**	14.0	12.0	**5.0**	8.1	7.4	9.1	18.7
28	HG-3SAT-V250-C1000-17	*highgirth*	3	250	1000	**6.0**	13.0	14.0	6.9	9.7	7.9	9.7	18.8
29	HG-3SAT-V250-C1000-18	*highgirth*	3	250	1000	**6.0**	10.0	11.0	**6.0**	9.1	7.4	9.3	17.7
30	HG-3SAT-V250-C1000-19	*highgirth*	3	250	1000	**5.0**	10.0	10.0	6.9	9.7	8.0	8.7	21.5
31	HG-3SAT-V250-C1000-20	*highgirth*	3	250	1000	**7.0**	10.0	13.0	**7.0**	9.3	8.0	9.3	19.0
32	HG-3SAT-V250-C1000-21	*highgirth*	3	250	1000	**4.0**	10.0	10.0	5.0	8.6	7.8	8.7	19.0
33	HG-3SAT-V250-C1000-22	*highgirth*	3	250	1000	**5.0**	16.0	11.0	**5.0**	9.2	7.5	7.9	18.0
34	HG-3SAT-V250-C1000-23	*highgirth*	3	250	1000	**6.0**	13.0	12.0	**6.0**	9.7	7.3	9.7	18.3

Continuation of Table 2 Average results of all algorithms tested on MSE 2020 and 2016 instances

	Instance	Group	n_l	n_x	n_c	SAT*Like*	SlsMcs	Cplex	AcoSat$^+_{\text{neg}}$	AcoSls$^+_{\text{neg}}$	Aco$^+_{\text{neg}}$	Aco$_{\text{neg}}$	Aco
35	HG-3SAT-V250-C1000-24	*highgirth*	3	250	1000	**5.0**	14.0	9.0	5.8	8.3	7.5	8.4	19.8
36	HG-3SAT-V250-C1000-100	*highgirth*	3	250	1000	**7.0**	12.0	14.0	**7.0**	9.6	7.9	9.5	19.3
37	HG-3SAT-V300-C1200-1	*highgirth*	3	300	1200	**7.0**	16.0	17.0	**7.0**	11.0	9.2	11.5	26.2
38	HG-3SAT-V300-C1200-2	*highgirth*	3	300	1200	6.0	12.0	13.0	**5.6**	11.9	9.4	12.2	30.1
39	HG-3SAT-V300-C1200-3	*highgirth*	3	300	1200	**6.0**	14.0	9.0	8.0	11.4	9.5	12.7	30.4
40	HG-3SAT-V300-C1200-4	*highgirth*	3	300	1200	**7.0**	12.0	13.0	**7.0**	10.7	9.5	11.8	28.3
41	HG-3SAT-V300-C1200-5	*highgirth*	3	300	1200	**6.0**	13.0	16.0	6.8	11.9	9.6	12.3	28.1
42	HG-3SAT-V300-C1200-6	*highgirth*	3	300	1200	**6.0**	15.0	14.0	**6.0**	11.5	8.4	12.8	25.6
43	HG-3SAT-V300-C1200-7	*highgirth*	3	300	1200	**5.0**	16.0	15.0	**5.0**	10.7	8.0	11.0	24.3
44	HG-3SAT-V300-C1200-8	*highgirth*	3	300	1200	**8.0**	16.0	14.0	**8.0**	12.1	10.0	12.8	28.7
45	HG-3SAT-V300-C1200-9	*highgirth*	3	300	1200	**7.0**	13.0	19.0	8.0	11.5	9.1	14.5	28.8
46	HG-3SAT-V300-C1200-10	*highgirth*	3	300	1200	**7.0**	12.0	13.0	**7.0**	11.0	9.5	12.4	29.4
47	HG-3SAT-V300-C1200-11	*highgirth*	3	300	1200	**6.0**	17.0	14.0	**6.0**	10.6	9.0	13.0	29.1
48	HG-3SAT-V300-C1200-12	*highgirth*	3	300	1200	**7.0**	13.0	14.0	**7.0**	12.0	9.7	11.7	28.1
49	HG-3SAT-V300-C1200-13	*highgirth*	3	300	1200	**7.0**	10.0	15.0	**7.0**	11.0	9.7	11.2	26.8
50	HG-3SAT-V300-C1200-14	*highgirth*	3	300	1200	**6.0**	12.0	14.0	6.1	9.5	8.9	10.7	25.1
51	HG-3SAT-V300-C1200-15	*highgirth*	3	300	1200	**7.0**	16.0	12.0	7.4	11.3	10.0	11.9	32.2
52	HG-3SAT-V300-C1200-16	*highgirth*	3	300	1200	**6.0**	17.0	14.0	**6.0**	11.4	9.4	11.8	26.3
53	HG-3SAT-V300-C1200-17	*highgirth*	3	300	1200	**7.0**	15.0	17.0	7.1	11.5	9.4	14.1	29.2
54	HG-3SAT-V300-C1200-18	*highgirth*	3	300	1200	**7.0**	11.0	13.0	7.4	10.4	9.0	11.3	27.0
55	HG-3SAT-V300-C1200-19	*highgirth*	3	300	1200	**6.0**	16.0	15.0	7.0	12.5	9.5	14.0	30.6
56	HG-3SAT-V300-C1200-20	*highgirth*	3	300	1200	**6.0**	14.0	15.0	**6.0**	11.1	9.7	12.5	27.8
57	HG-3SAT-V300-C1200-21	*highgirth*	3	300	1200	6.0	9.0	16.0	**5.0**	9.7	9.3	11.2	27.3
58	HG-3SAT-V300-C1200-22	*highgirth*	3	300	1200	**6.0**	17.0	18.0	**6.0**	11.1	9.4	12.4	29.1
59	HG-3SAT-V300-C1200-23	*highgirth*	3	300	1200	**7.0**	12.0	15.0	**7.0**	10.7	8.7	11.1	27.0
60	HG-3SAT-V300-C1200-24	*highgirth*	3	300	1200	**7.0**	12.0	18.0	**7.0**	10.9	9.1	12.8	28.1
61	HG-3SAT-V300-C1200-100	*highgirth*	3	300	1200	**7.0**	14.0	14.0	**7.0**	11.3	9.1	11.4	29.8
62	HG-4SAT-V100-C900-2	*highgirth*	4	100	900	**2.0**	4.0	4.0	**2.0**	3.6	**2.0**	2.9	3.9
63	HG-4SAT-V100-C900-4	*highgirth*	4	100	900	**2.0**	4.0	3.0	**2.0**	3.2	2.7	3.6	5.1
64	HG-4SAT-V100-C900-7	*highgirth*	4	100	900	**2.0**	6.0	3.0	**2.0**	3.6	2.9	3.6	4.5
65	HG-4SAT-V100-C900-14	*highgirth*	4	100	900	**2.0**	6.0	4.0	**2.0**	3.0	2.3	3.9	4.3
66	HG-4SAT-V100-C900-19	*highgirth*	4	100	900	**2.0**	4.0	4.0	2.6	3.1	2.6	3.2	4.5
67	HG-4SAT-V100-C900-20	*highgirth*	4	100	900	**2.0**	5.0	**2.0**	**2.0**	2.7	2.3	2.8	4.2
68	HG-4SAT-V100-C900-23	*highgirth*	4	100	900	**2.0**	3.0	4.0	**2.0**	2.8	**2.0**	2.6	4.1

Continuation of Table 2 Average results of all algorithms tested on MSE 2020 and 2016 instances

	Instance	Group	n_l	n_x	n_c	SAT*Like*	SlsMcs	Cplex	AcoSat$^{+}_{\text{neg}}$	AcoSls$^{+}_{\text{neg}}$	Aco$^{+}_{\text{neg}}$	Aco$_{\text{neg}}$	Aco
69	HG-4SAT-V150-C1350-1	*highgirth*	4	150	1350	**1.0**	5.0	8.0	**1.0**	4.7	2.9	6.1	8.1
70	HG-4SAT-V150-C1350-2	*highgirth*	4	150	1350	**2.0**	7.0	8.0	**2.0**	4.9	3.3	6.8	9.9
71	HG-4SAT-V150-C1350-3	*highgirth*	4	150	1350	2.0	6.0	6.0	1.9	2.8	**1.3**	1.8	6.3
72	HG-4SAT-V150-C1350-4	*highgirth*	4	150	1350	**2.0**	5.0	7.0	**2.0**	3.6	3.0	3.6	8.4
73	HG-4SAT-V150-C1350-5	*highgirth*	4	150	1350	**2.0**	7.0	7.0	**2.0**	4.2	3.2	4.3	9.0
74	HG-4SAT-V150-C1350-6	*highgirth*	4	150	1350	**1.0**	9.0	6.0	2.0	4.3	2.4	5.8	9.0
75	HG-4SAT-V150-C1350-7	*highgirth*	4	150	1350	**2.0**	5.0	7.0	**2.0**	4.2	3.7	3.9	8.5
76	HG-4SAT-V150-C1350-8	*highgirth*	4	150	1350	**2.0**	8.0	10.0	**2.0**	4.5	3.7	5.2	10.4
77	HG-4SAT-V150-C1350-9	*highgirth*	4	150	1350	**2.0**	7.0	8.0	**2.0**	4.6	3.3	4.7	10.2
78	HG-4SAT-V150-C1350-10	*highgirth*	4	150	1350	**2.0**	8.0	7.0	2.4	4.6	3.5	7.0	8.9
79	HG-4SAT-V150-C1350-11	*highgirth*	4	150	1350	**2.0**	4.0	5.0	2.7	4.1	3.1	4.1	8.2
80	HG-4SAT-V150-C1350-12	*highgirth*	4	150	1350	**2.0**	6.0	10.0	**2.0**	3.7	3.1	4.3	7.6
81	HG-4SAT-V150-C1350-13	*highgirth*	4	150	1350	**2.0**	7.0	8.0	2.4	4.6	4.0	6.5	8.1
82	HG-4SAT-V150-C1350-14	*highgirth*	4	150	1350	**2.0**	6.0	7.0	**2.0**	4.4	2.8	3.9	8.4
83	HG-4SAT-V150-C1350-15	*highgirth*	4	150	1350	**2.0**	5.0	6.0	**2.0**	3.7	2.6	4.5	9.4
84	HG-4SAT-V150-C1350-16	*highgirth*	4	150	1350	**2.0**	6.0	**2.0**	2.4	5.0	3.4	7.5	9.8
85	HG-4SAT-V150-C1350-17	*highgirth*	4	150	1350	**2.0**	7.0	4.0	**2.0**	4.5	3.6	4.2	9.1
86	HG-4SAT-V150-C1350-18	*highgirth*	4	150	1350	**2.0**	6.0	7.0	2.7	4.6	3.0	3.7	8.9
87	HG-4SAT-V150-C1350-19	*highgirth*	4	150	1350	2.0	9.0	4.0	**1.9**	2.6	2.4	2.8	9.2
88	HG-4SAT-V150-C1350-20	*highgirth*	4	150	1350	**2.0**	8.0	10.0	2.5	4.9	2.8	8.0	8.0
89	HG-4SAT-V150-C1350-21	*highgirth*	4	150	1350	**2.0**	6.0	5.0	**2.0**	4.4	3.4	5.8	9.4
90	HG-4SAT-V150-C1350-22	*highgirth*	4	150	1350	**2.0**	8.0	7.0	2.6	4.5	2.7	3.6	8.4
91	HG-4SAT-V150-C1350-23	*highgirth*	4	150	1350	**2.0**	6.0	7.0	**2.0**	5.0	4.1	8.6	9.6
92	HG-4SAT-V150-C1350-24	*highgirth*	4	150	1350	**2.0**	6.0	5.0	**2.0**	4.9	3.2	7.1	8.8
93	HG-4SAT-V150-C1350-100	*highgirth*	4	150	1350	**2.0**	5.0	8.0	**2.0**	5.0	3.6	9.0	8.5
94	scpcyc06_maxsat	$setcov_1$	4	192	432	**60.0**	**60.0**	**60.0**	**60.0**	60.3	**60.0**	95.6	94.5
95	scpcyc07_maxsat	$setcov_1$	4	448	1120	158.0	**148.0**	152.0	158.3	149.7	151.0	274.9	274.0
96	scpcyc08_maxsat	$setcov_1$	4	1024	2816	392.0	**361.0**	370.0	391.4	361.5	361.7	735.3	735.2
97	scpcyc09_maxsat	$setcov_2$	4	2304	6912	836.0	**815.0**	924.0	824.5	830.7	963.9	1895.9	1872.6
98	scpcyc10_maxsat	$setcov_2$	4	5120	16640	1922.0	1917.0	16640.0	1932.4	**1916.3**	2264.6	4630.2	4615.9
99	scpcyc11_maxsat	$setcov_2$	4	11264	39424	4339.0	**4295.0**	39424.0	11042.9	11003.6	5317.4	11015.8	11040.9
100	ram_k3_n9.ra0	$ramsey_1$	6	36	210	**1.0**	**1.0**	**1.0**	**1.0**	**1.0**	**1.0**	**1.0**	**1.0**
101	ram_k3_n11.ra0	$ramsey_1$	6	55	495	**7.0**	**7.0**	**7.0**	**7.0**	**7.0**	**7.0**	**7.0**	**7.0**
102	ram_k3_n12.ra0	$ramsey_1$	6	66	715	**10.0**	**10.0**	11.0	**10.0**	**10.0**	**10.0**	10.3	**10.0**

Continuation of Table 2 Average results of all algorithms tested on MSE 2020 and 2016 instances

	Instance	Group	n_l	n_x	n_c	SAT*Like*	SlsMcs	Cplex	$\text{AcoSat}^+_{\text{neg}}$	$\text{AcoSls}^+_{\text{neg}}$	$\text{Aco}^+_{\text{neg}}$	Aco_{neg}	Aco
103	ram_k3_n14.ra0	$ramsey_1$	6	91	1365	**21.0**	**21.0**	**21.0**	**21.0**	**21.0**	21.7	22.6	22.7
104	ram_k3_n15.ra0	$ramsey_2$	6	105	1820	**30.0**	**30.0**	31.0	**30.0**	**30.0**	30.6	**30.0**	**30.0**
105	ram_k3_n16.ra0	$ramsey_2$	6	120	2380	**39.0**	**39.0**	42.0	**39.0**	**39.0**	40.0	40.9	40.1
106	ram_k3_n17.ra0	$ramsey_2$	6	136	3060	**50.0**	**50.0**	54.0	**50.0**	**50.0**	51.5	51.4	51.7
107	ram_k3_n18.ra0	$ramsey_2$	6	153	3876	**60.0**	**60.0**	70.0	**60.0**	**60.0**	64.3	69.3	66.6
108	ram_k4_n18.ra0	$ramsey_2$	6	153	6120	**9.0**	**9.0**	14.0	**9.0**	**9.0**	10.9	17.3	19.0
109	ram_k3_n19.ra0	$ramsey_2$	6	171	4845	**75.0**	**75.0**	87.0	**75.0**	**75.0**	79.8	89.9	84.3
110	ram_k4_n19.ra0	$ramsey_2$	6	171	7752	**15.0**	**15.0**	29.0	**15.0**	**15.0**	18.6	33.5	33.3
111	scpclr10_maxsat	$setcov_3$	126	210	721	**25.0**	**25.0**	**25.0**	**25.0**	25.3	**25.0**	46.6	45.9
112	scpclr12_maxsat	$setcov_3$	330	495	2542	28.0	**23.0**	24.0	27.0	26.7	23.6	80.6	80.9
113	scpclr13_maxsat	$setcov_3$	495	715	4810	30.0	28.0	31.0	29.5	**26.8**	28.5	123.8	122.8